

TradeAlerts Project Overview

■ TradeAlerts Project: HighLevel Architecture

1. Purpose & Workflow

- Signal Scanner: Runs continuously (every 60s) over your S&P500 universe, fetching raw price quotes via the E*TRADE API and intraday statistics (RSI, MACD, Bollinger Bands, VWAP, volume) via Yahoo Finance. It applies your configured thresholds to identify Buy and Sell signals.

- Alert Delivery: When a signal meets your confidence criteria, the scanner pings the Dashboard backend via a POST to /api/alerts. Alerts are recorded in your local alerts_clean.db SQLite table, then surfaced in the Trade Alerts page.

- Dashboard UI: A Flask app with four blueprints:

1. API Blueprint (/api): receives and exposes alerts JSON.

2. Alerts Blueprint (/alerts): renders the alerts table with SVG sparklines and Clear All.

3. Simulation Blueprint (/simulation): allows simulating buys at alert time, tracking cash/positions/trade history in simulation.db, and resetting the simulation.

4. Status Blueprint (/status): shows live connectivity indicators for Yahoo Finance Intraday and E*TRADE API with a one-click Reconnect API link.

- Styling & Layout: A single base.html using Bootstrap's container-fluid and a darkmode theme. All pages extend base.html for consistent headers, typography, and responsive behavior. The Alerts page spans the full window, with centered title and filters, right-aligned status badges, and a horizontally scrollable, no-wrap table with SVG sparklines. The Simulation page mirrors this darktheme aesthetic, with a Reset button, realtime cash/open value/realized P&L stats, and tables for open positions and trade history.

■ Project Structure

TradeAlerts/

■ dashboard.py

■ alerts_clean.db

■ simulation.db

■ templates/

■ ■ base.html

■ ■ alerts.html

■ ■ simulation/index.html

■ static/

■ ■ style.css

■ blueprints/

■ ■ api/views.py

■ ■ alerts/views.py

■ ■ simulation/views.py

■ ■ status/views.py

■ services/

■ ■ alert_service.py

■ ■ simulation_service.py

■ ■ status_service.py

■ Key Components & Recent Enhancements

Component	Role	Recent Tweaks
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Scanner (scanner.py)	Fetches raw price from E*TRADE API and computes intraday indicators via Yahoo Finance → POST /api/alerts	Unified
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MACD/RSI/VWAP/Volume logic, fixed formatting
alert_service.py CRUD for alerts + sparkline SVG generator Inline
SVG sparklines, trigger parsing, clean filtering
alerts.html Main Trade Alerts page Bootstrap table, full-width layout,
centered filters, stacked status badges
simulation_service.py Manages cash, positions, trade history Buy/reset
endpoints with stub logic
simulation/index.html Simulation Dashboard Dark-mode tables for positions
and history
status_service.py Checks connectivity to Yahoo & E*TRADE Live status
badges, reconnect link

■ Where We Stand

- All core flows (price fetch, indicator calc, alert ingestion, display, filtering, simulation) are wired end-to-end.
- UI consistency via base.html + Bootstrap.
- Interactive features (filters, clear, simulation buy/reset) operational.
- SVG sparklines provide quick trend views.

Next steps may include:

- Historical backtesting
- Notification hooks (email, Slack)
- Expanded indicator set (ATR, stochastic, AI-driven)
- Responsive refinements (mobile interactivity)

Full Scanner Simulation

Our full scanner simulation engine iterates through the entire symbol universe, applies all enabled indicator filters (SMA, RSI, MACD histogram, Bollinger Bands breakout, VWAP threshold, etc.), and simulates trades in real-time or historical replay mode. It logs buy/sell events, reasons for decisions, and portfolio performance over time.

Full Scanner Backtesting

The backtesting module replays historical 1-minute and 1-day bar data across the full symbol set, applying the same indicator logic and trade execution rules. It calculates detailed metrics including win rate, average return, maximum drawdown, and Sharpe ratio—enabling side-by-side comparison of strategy variations.

Complex Indicator Logic

We support a full suite of technical indicators:

- **Simple Moving Average (SMA):** Configurable lookback and cross-over triggers.
- **Relative Strength Index (RSI):** Overbought/oversold thresholds and slope detection.
- **MACD Histogram:** Fast/slow EMA spans and signal line for momentum.
- **Bollinger Bands:** Customizable length and standard-deviation multipliers for breakout detection.
- **Volume Weighted Average Price (VWAP):** Intraday benchmark with threshold offsets.
- **Custom Combinations:** Flexible toggles for mixing any subset of the above filters in any order.